Augmented Reality (AR)
Working of AR

Introduction

- **Augmented Reality (AR)** enhances the real world by overlaying digital content (images, 3D objects, text, sound) onto physical environment in real time.
- AR platforms
  - **Google ARCore**
  - **Apple ARKit**

Step-by-Step Working of AR

- Augmented Reality (AR) functions through a continuous loop of below steps
  - Scene Capture
  - Scene Understanding
  - Tracking
  - Rendering Digital Content
  - Interaction

Scene Capture

- A device (smartphone, tablet, AR glasses) uses:
  - **Camera**
  - **GPS**
  - **Accelerometer**
  - **Gyroscope**
  - **Depth sensors (LiDAR in advanced devices)**
- These sensors capture the real-world environment.

Scene Understanding

- The AR system analyzes:
- Surfaces (floors, tables, walls)
- Light conditions
- Object positions
- Distance and depth
- This process is called **environment mapping** or **SLAM (Simultaneous Localization and Mapping)**.

Tracking

- AR tracks the device's position and orientation in real time using:
  - **Marker-based tracking**
    - Uses QR codes, images
  - **Marker-less tracking**
    - Uses surface detection
  - **GPS-based tracking**
    - Uses GPS
  - **Projection-Based AR**
    - Projects light onto surfaces

Rendering Digital Content

- The system places 3D models, animations, or text in the correct position using:
  - Computer vision
  - 3D graphics engine
  - Real-time rendering
- Example:
  - In **Pokémon Go**, Pokémon appear on your street through your phone camera.

Interaction

- Users can:
  - Touch the screen
  - Move the device
  - Use gestures or voice commands
- The AR system updates the digital objects instantly based on user movement.

Thank You…!!!